ESP Flash Tool

UML Diagrams

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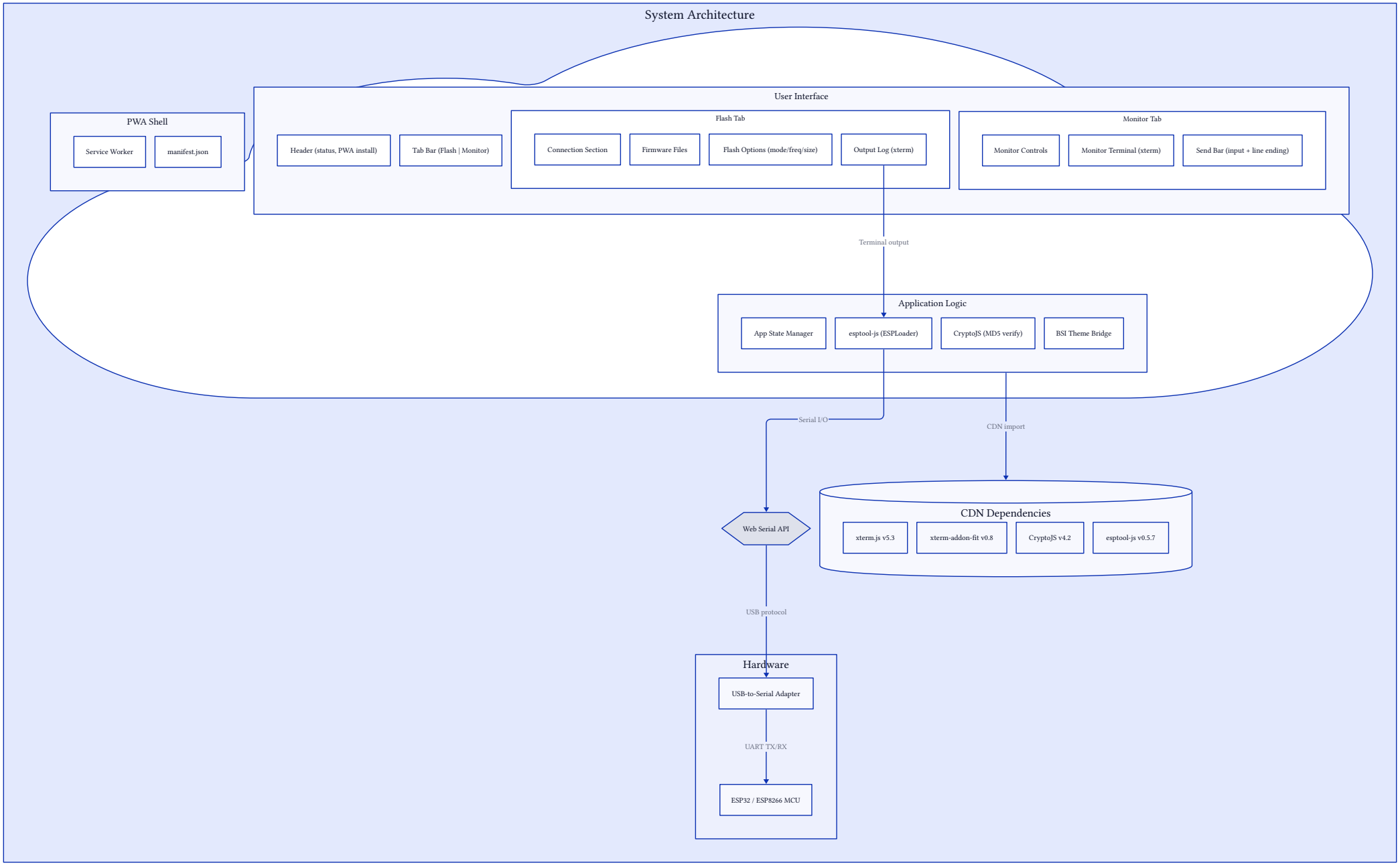
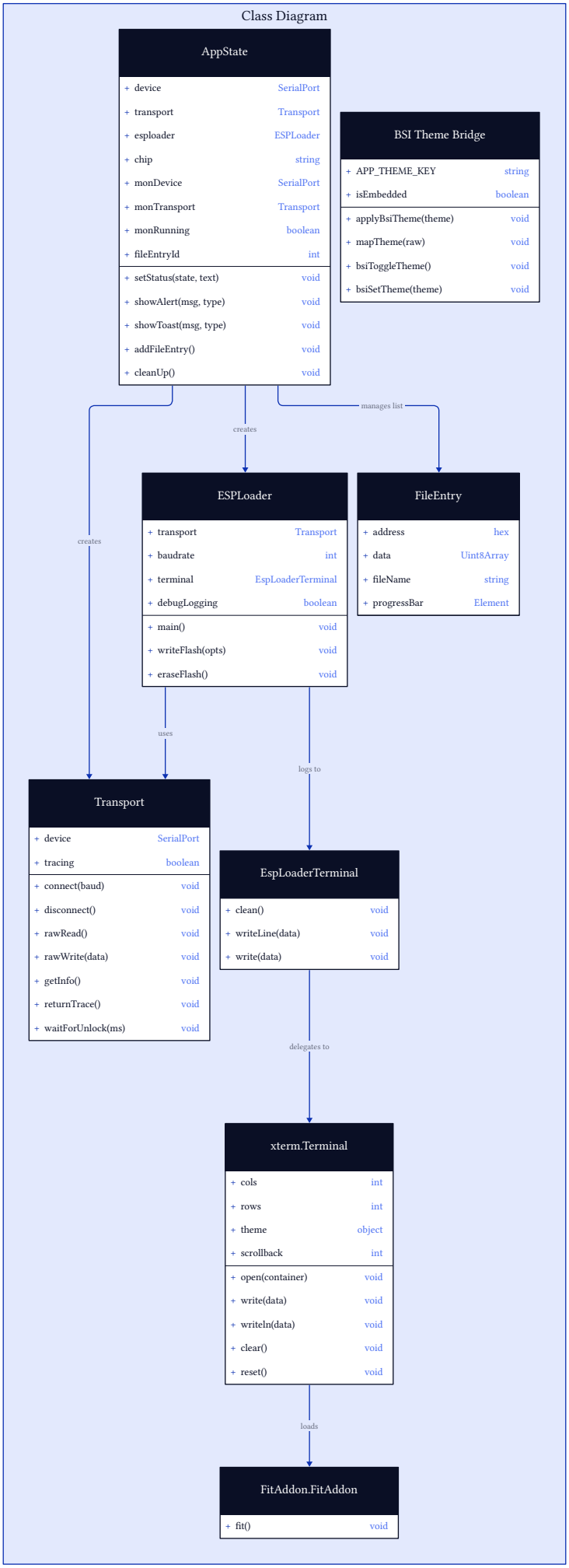
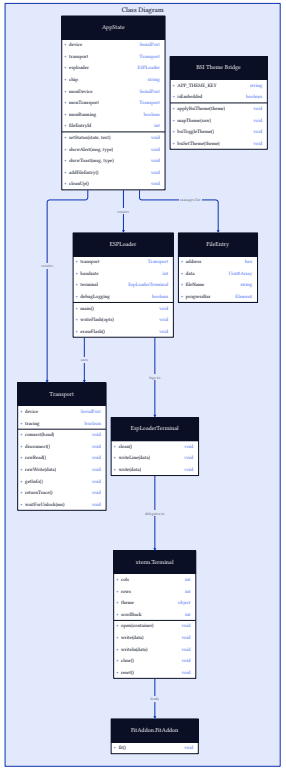
Architecture Diagram

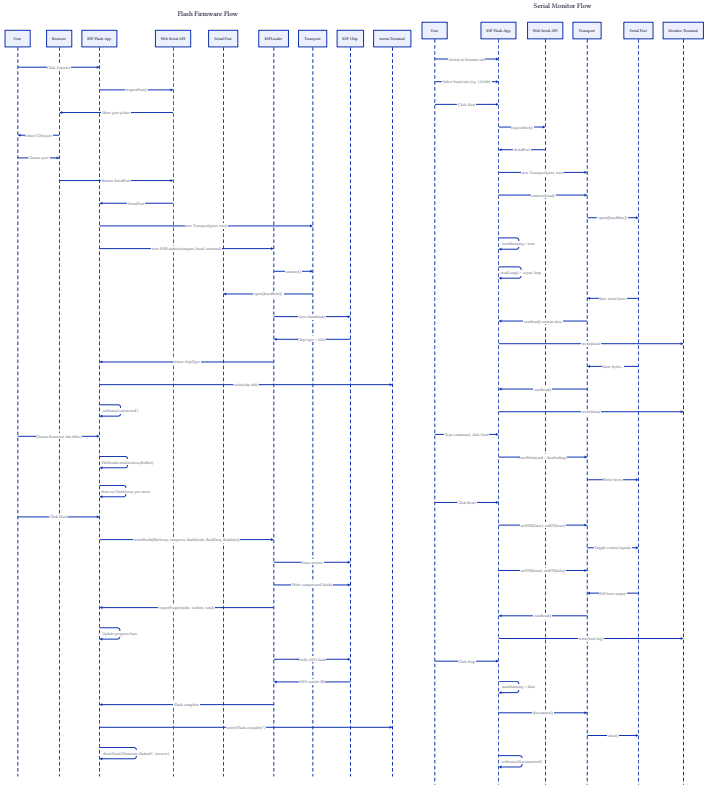


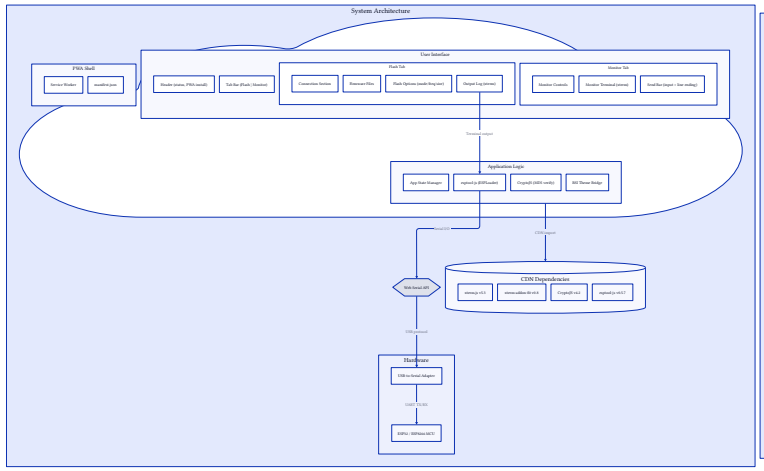
Class Diagram

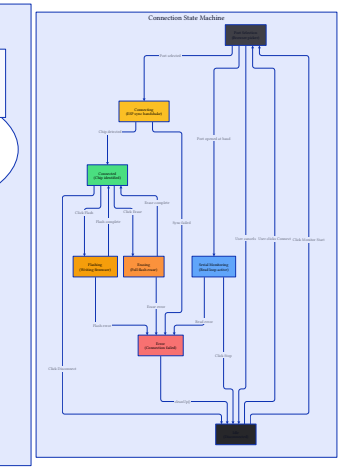


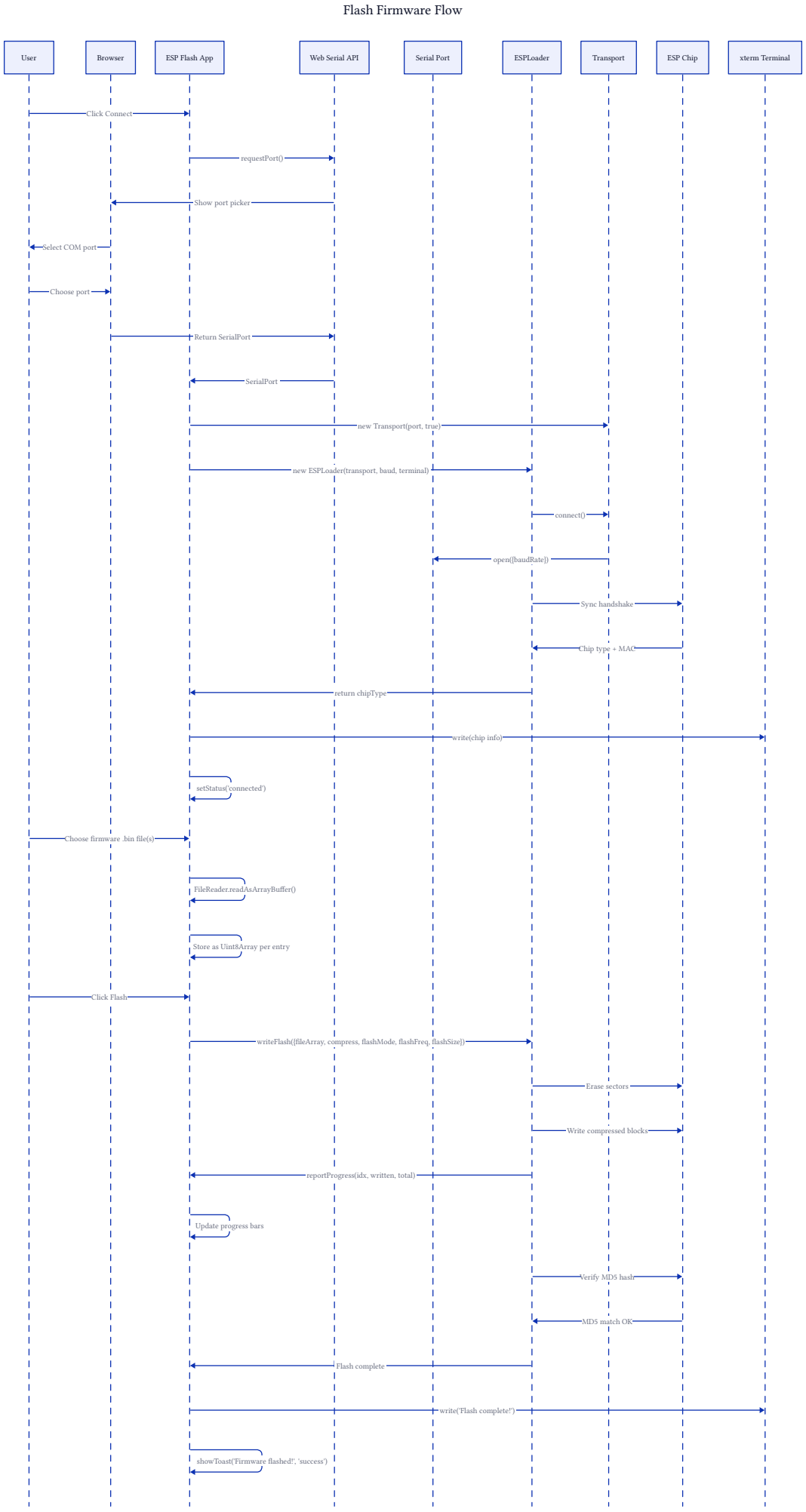
Combined UML Diagrams



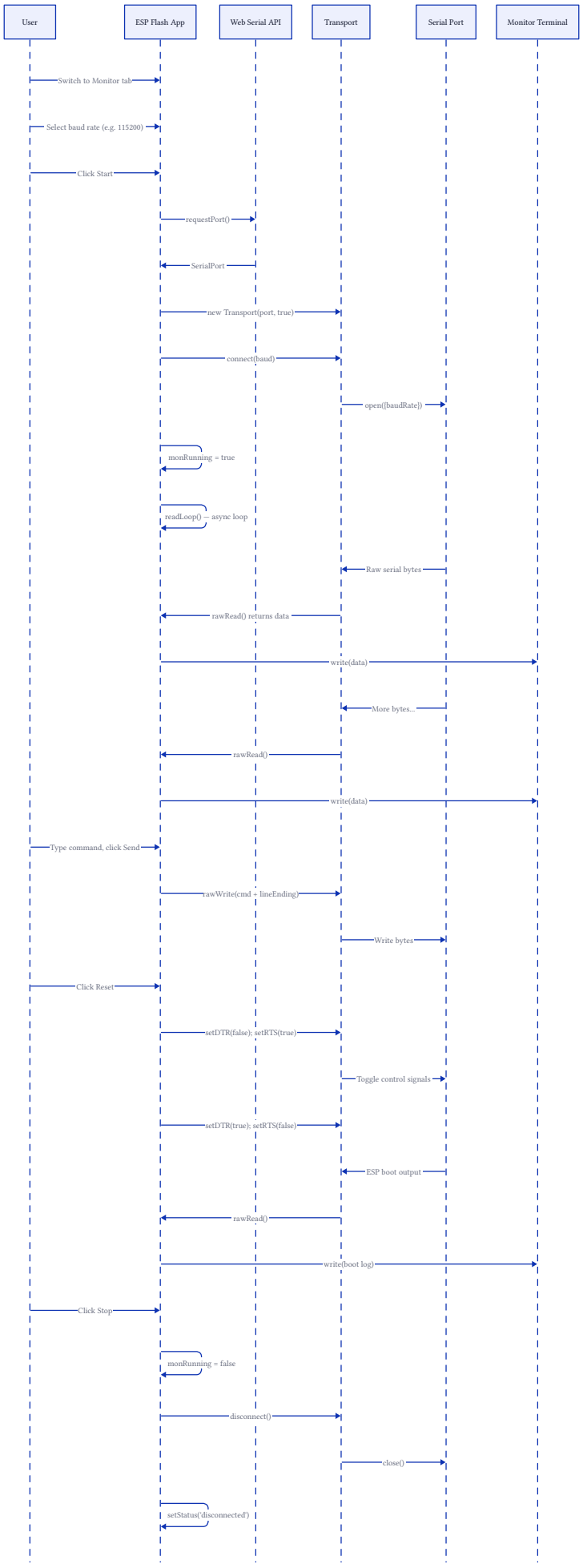


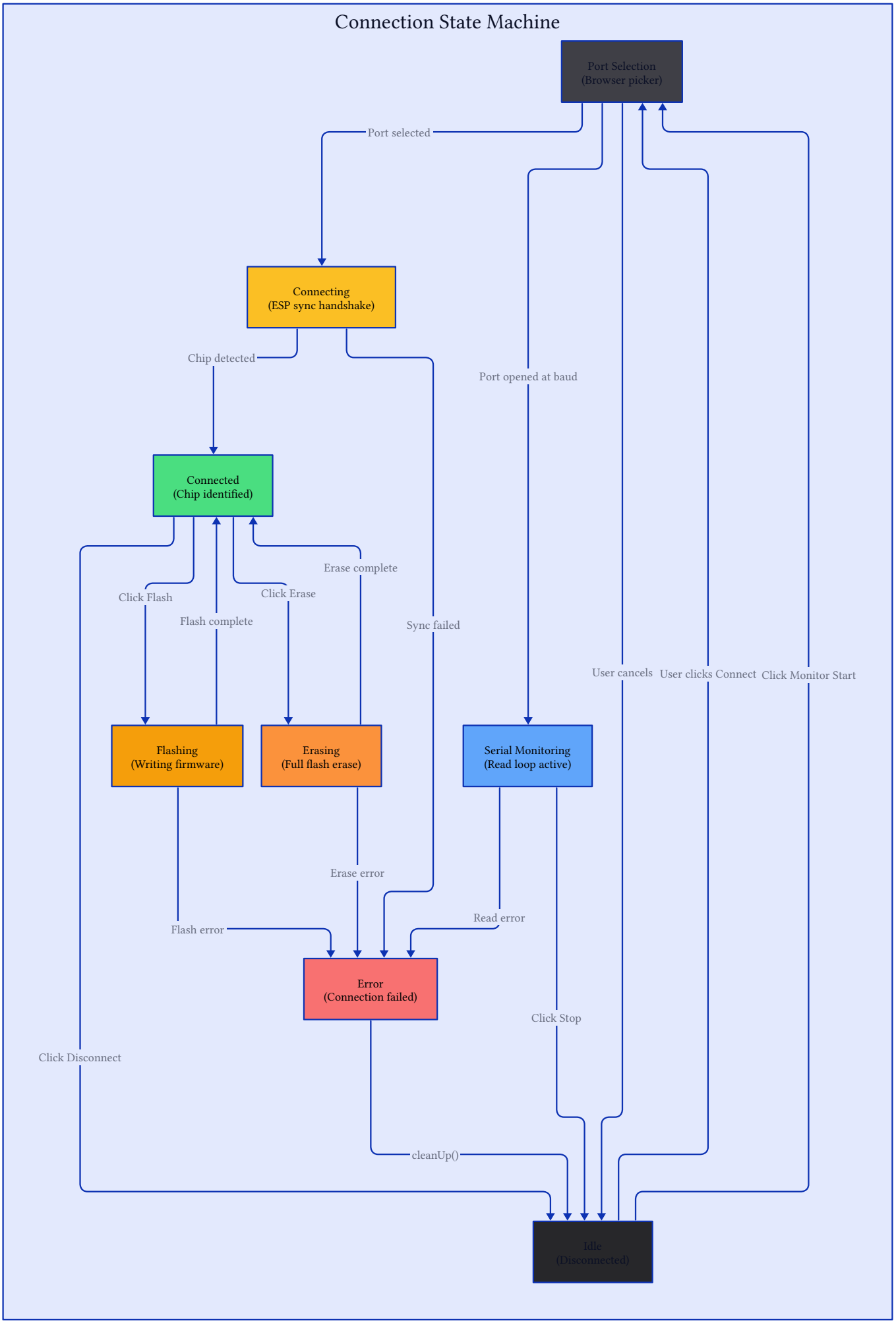



Connection State Machine
Disconnected
(idle, no port)
Port selected
Connecting
(ESP Loader handshake)
Chip detected
Port opened at baud
Connected
(Chip ready)
Chip Flash
Chip Erase
Read complete
Flash complete
Erase complete
Sync error
Flashing
(Writing firmware)
Erasing
(Full flash erase)
Serial Monitoring
(Read loop active)
Error
(Connection failed)
Flash error
Erase error
Sync error
Chip Disconnect
Chip Reset
User unsels / Port dies / Connect
Click Monitor Start
Chip Disconnect
Port closed
Port closed
Port closed


Flash Firmware Flow
User
Browser
ESP Flash App
Web Serial API
Serial Port
ESPLoader
Transport
ESP Chip
xterm Terminal
Click Connect
requestPort()
Show port picker
Select COM port
Choose port
Return SerialPort
SerialPort
new Transport(port, true)
new ESPLoader(transport, baud, terminal)
connect()
open({baudRate})
Sync handshake
Chip type + MAC
return chipType
write(chip info)
setStatus('connected')
Choose firmware .bin file(s)
FileReader.readAsArrayBuffer()
Store as Uint8Array per entry
Click Flash
writeFlash({fileArray, compress, flashMode, flashFreq, flashSize})
Erase sectors
Write compressed blocks
reportProgress(idx, written, total)
Update progress bars
Verify MD5 hash
MD5 match OK
Flash complete
write('Flash complete!')
showToast('Firmware flashed!', 'success')

Serial Monitor Flow




Connection State Machine
Port Selection
(Browser picker)
Port selected
Connecting
(ESP sync handshake)
Chip detected
Port opened at baud
Connected
(Chip identified)
Erase complete
Click Flash
Click Erase
Flash complete
Sync failed
User cancels
User clicks Connect
Click Monitor Start
Flashing
(Writing firmware)
Erasing
(Full flash erase)
Serial Monitoring
(Read loop active)
Erase error
Read error
Flash error
Error
(Connection failed)
Click Stop
Click Disconnect
cleanUp()
Idle
(Disconnected)